Visualization Methods in Data Analysis by R

Laboratory Works Portfolio – Group 4, Variant 5

Group 4

2026-03-09

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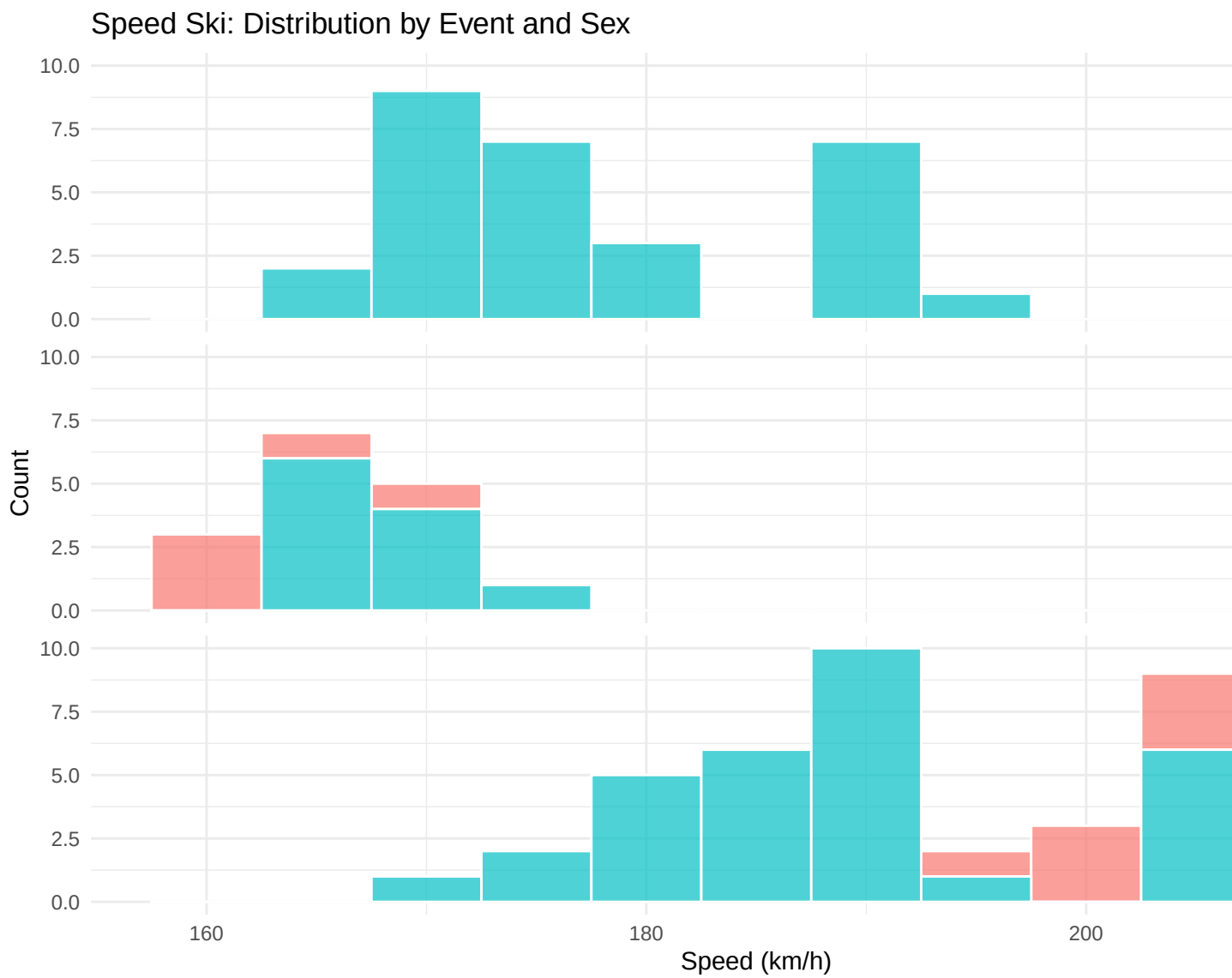
Introduction

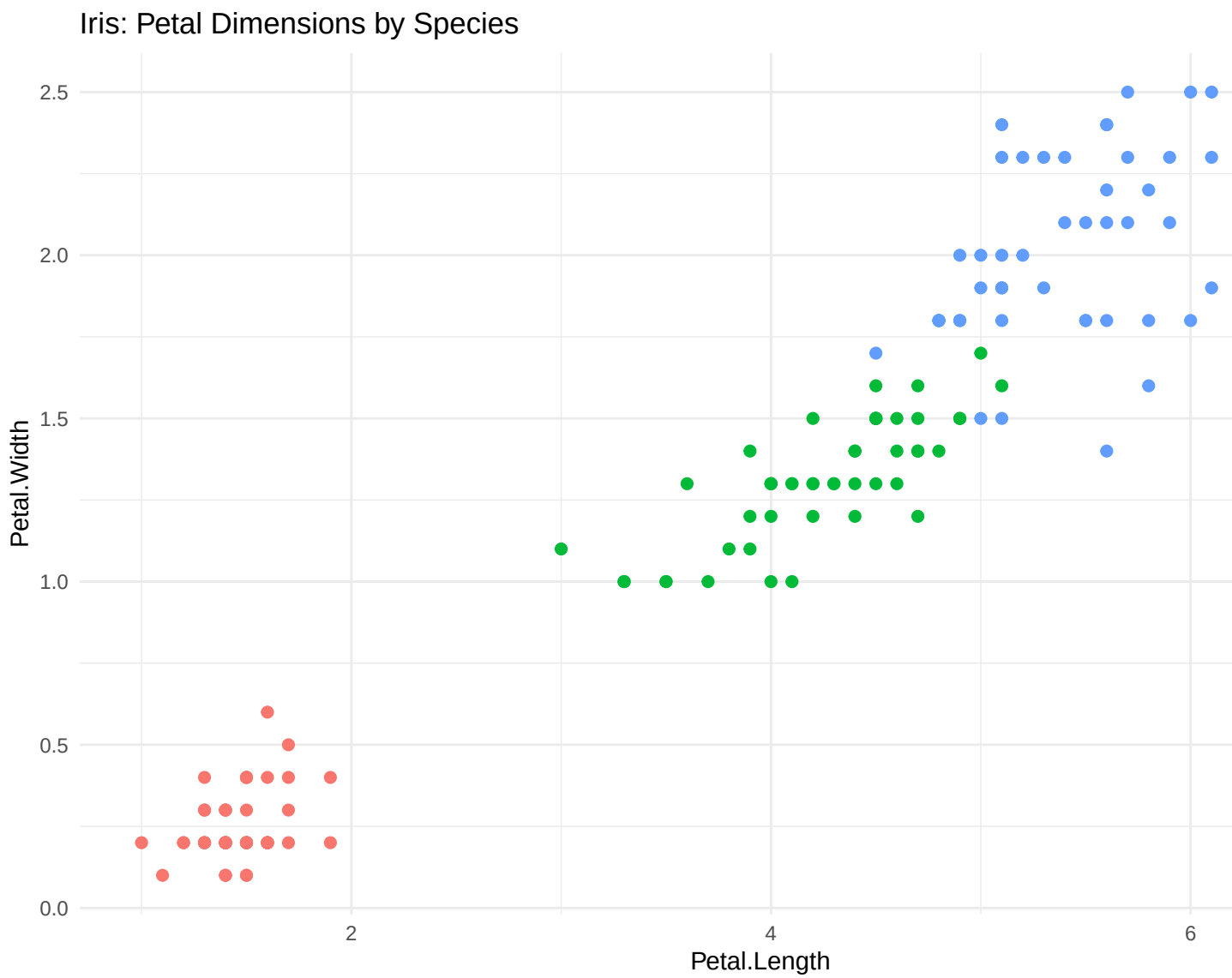
This report is the final portfolio for the course "Visualization Methods in Data Analysis by R" (2026). It contains highlights from all laboratory works (Lab 0–9).

Lab 0: R Setup and Basics

```r
library(GDAdata)

ggplot(SpeedSki, aes(Speed, fill = Sex)) +
  geom_histogram(binwidth = 5, color = "white", alpha = 0.7) +
  facet_grid(rows = vars(Event)) +
  labs(title = "Speed Ski: Distribution by Event and Sex",
       x = "Speed (km/h)", y = "Count") +
  theme_minimal()
```

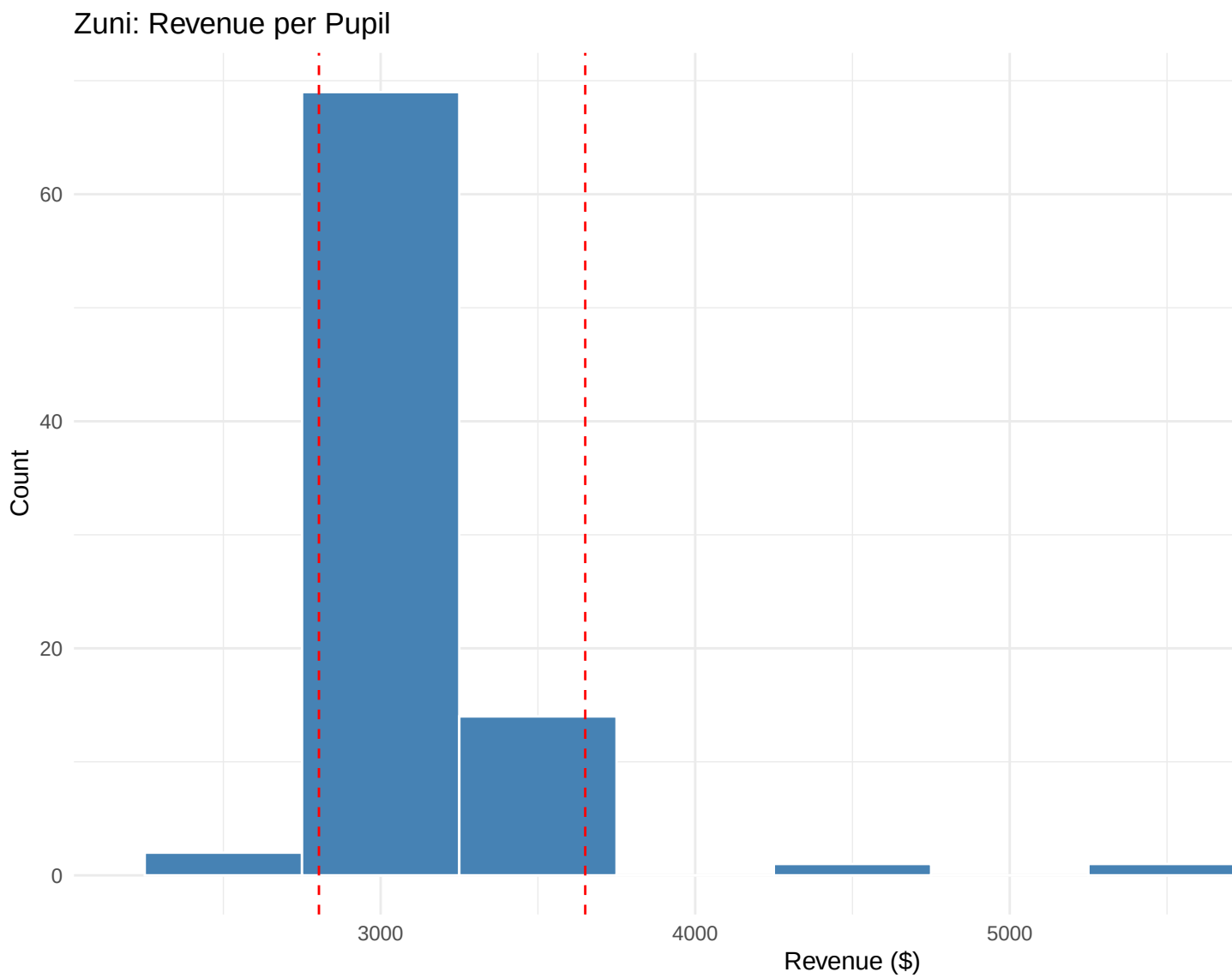



```r
ggplot(iris, aes(Petal.Length, Petal.Width, colour = Species)) +
  geom_point(size = 2) +
  labs(title = "Iris: Petal Dimensions by Species") +
  theme_minimal()
```

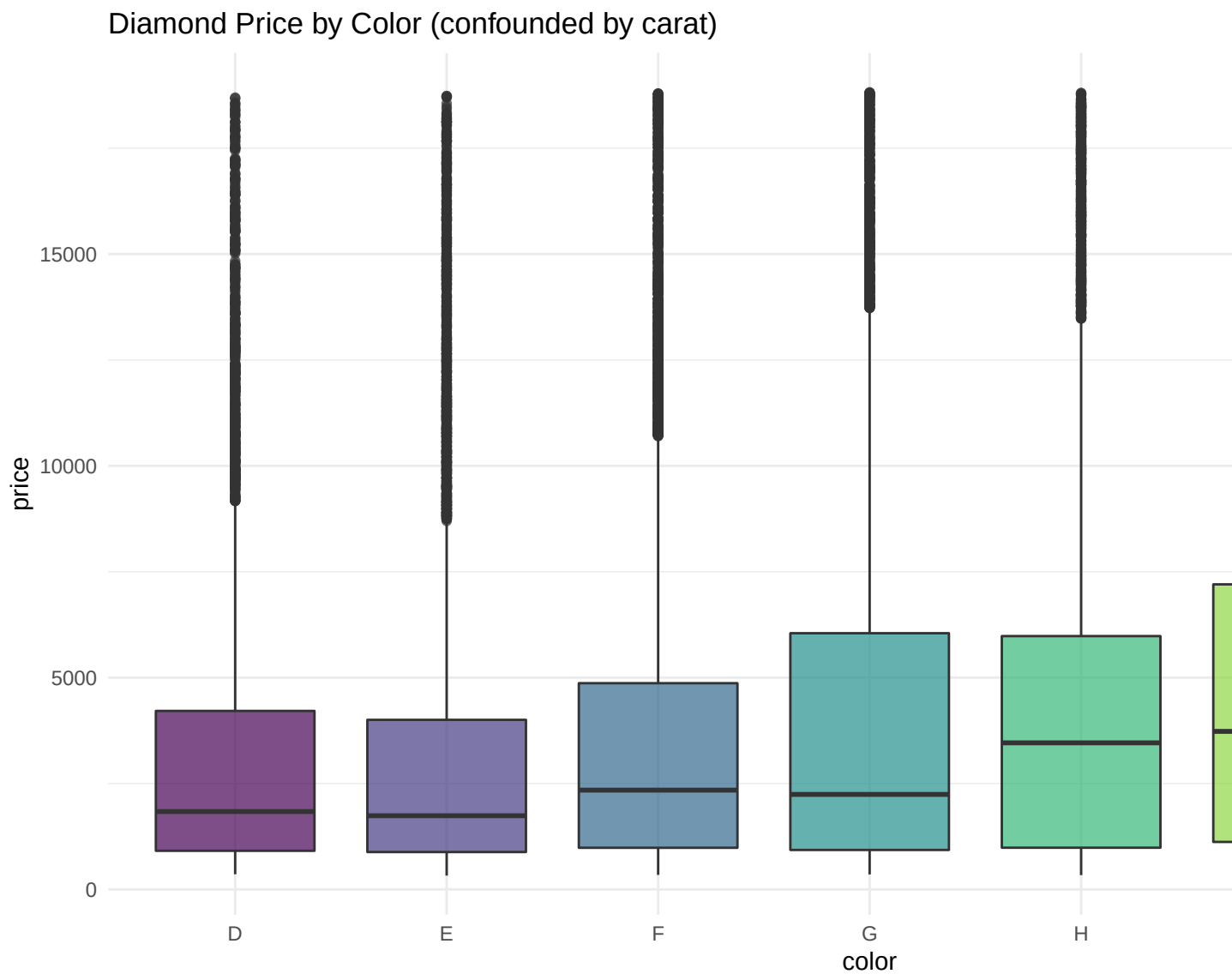


Lab 1: Exercises from GDA with R

```
library(lawstat)
data(zuni)

q05 <- quantile(zuni$Revenue, 0.05)
q95 <- quantile(zuni$Revenue, 0.95)

ggplot(zuni, aes(x = Revenue)) +
  geom_histogram(binwidth = 500, fill = "steelblue", color = "white") +
  geom_vline(xintercept = c(q05, q95), color = "red", linetype = "dashed") +
  labs(title = "Zuni: Revenue per Pupil", x = "Revenue ($)", y = "Count") +
  theme_minimal()
```

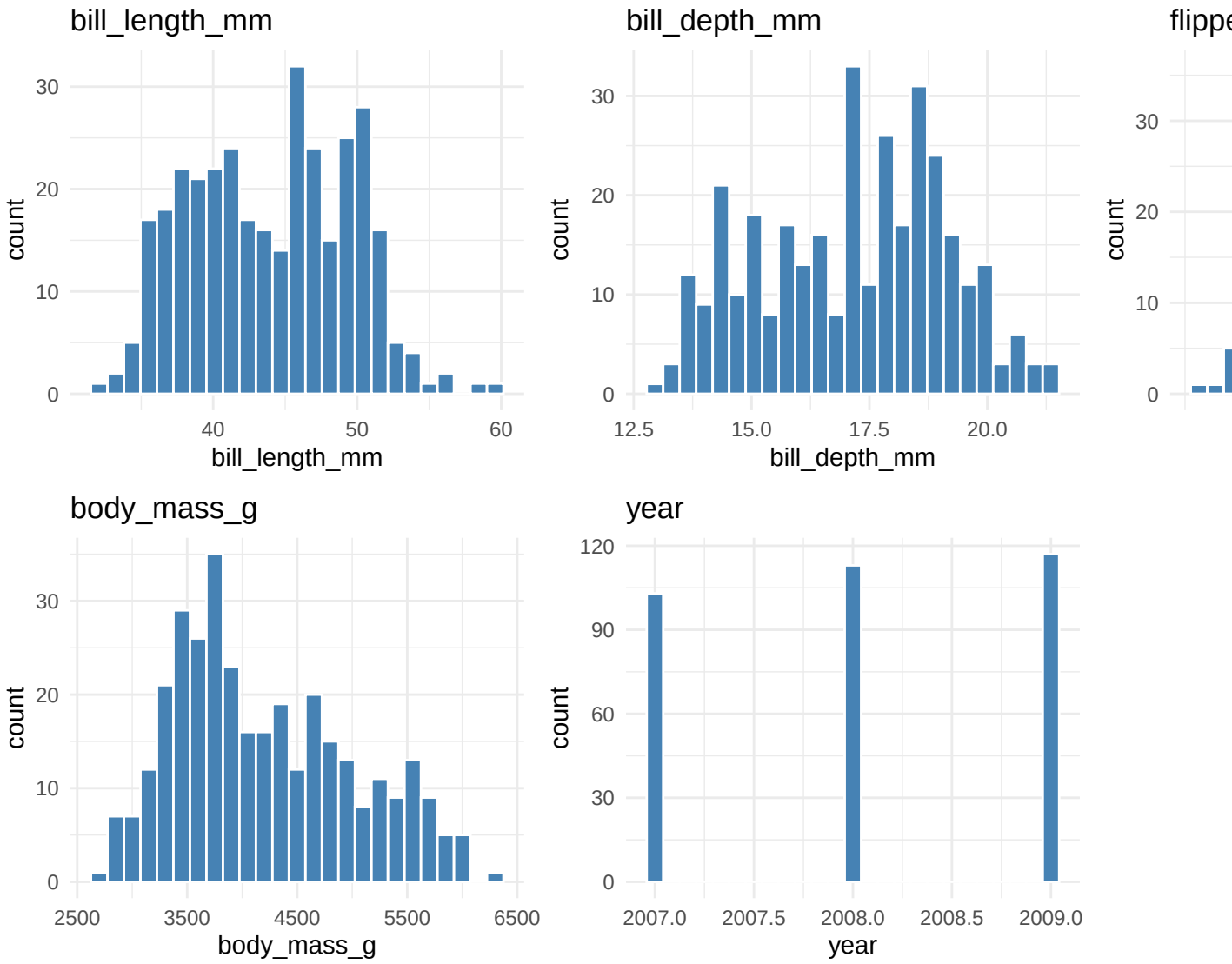



```r
ggplot(diamonds, aes(x = color, y = price, fill = color)) +
  geom_boxplot(alpha = 0.7, show.legend = FALSE) +
  scale_fill_viridis_d() +
  labs(title = "Diamond Price by Color (confounded by carat)") +
  theme_minimal()
```



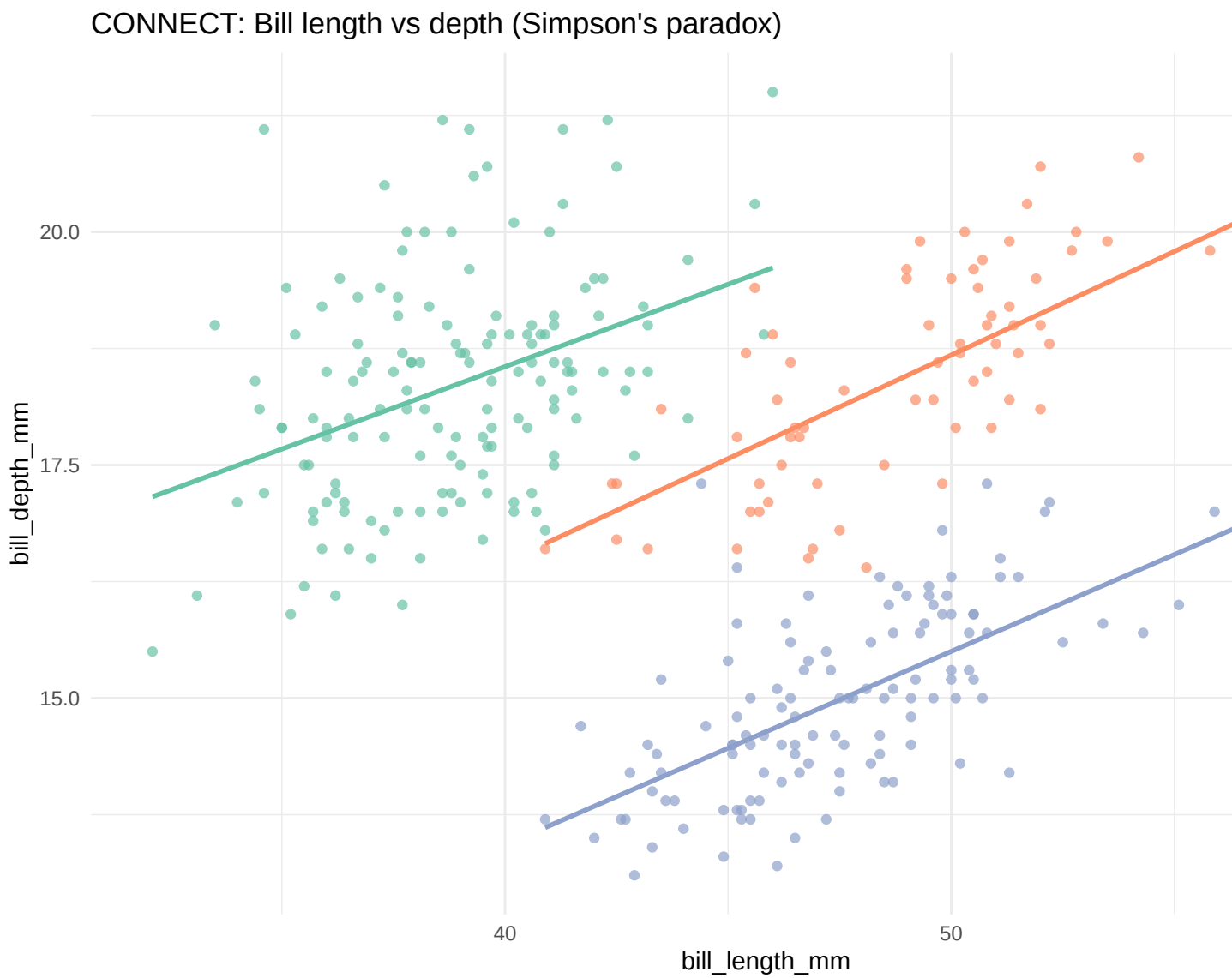
Lab 2: autoGEDA

```
data(penguins)
penguins_clean <- penguins |> drop_na()

numeric_vars <- penguins_clean |> select(where(is.numeric)) |> names()

plots <- map(numeric_vars, function(var) {
  ggplot(penguins_clean, aes(x = .data[[var]])) +
    geom_histogram(fill = "steelblue", color = "white", bins = 25) +
    labs(title = var) + theme_minimal()
})

wrap_plots(plots, ncol = 3)
```

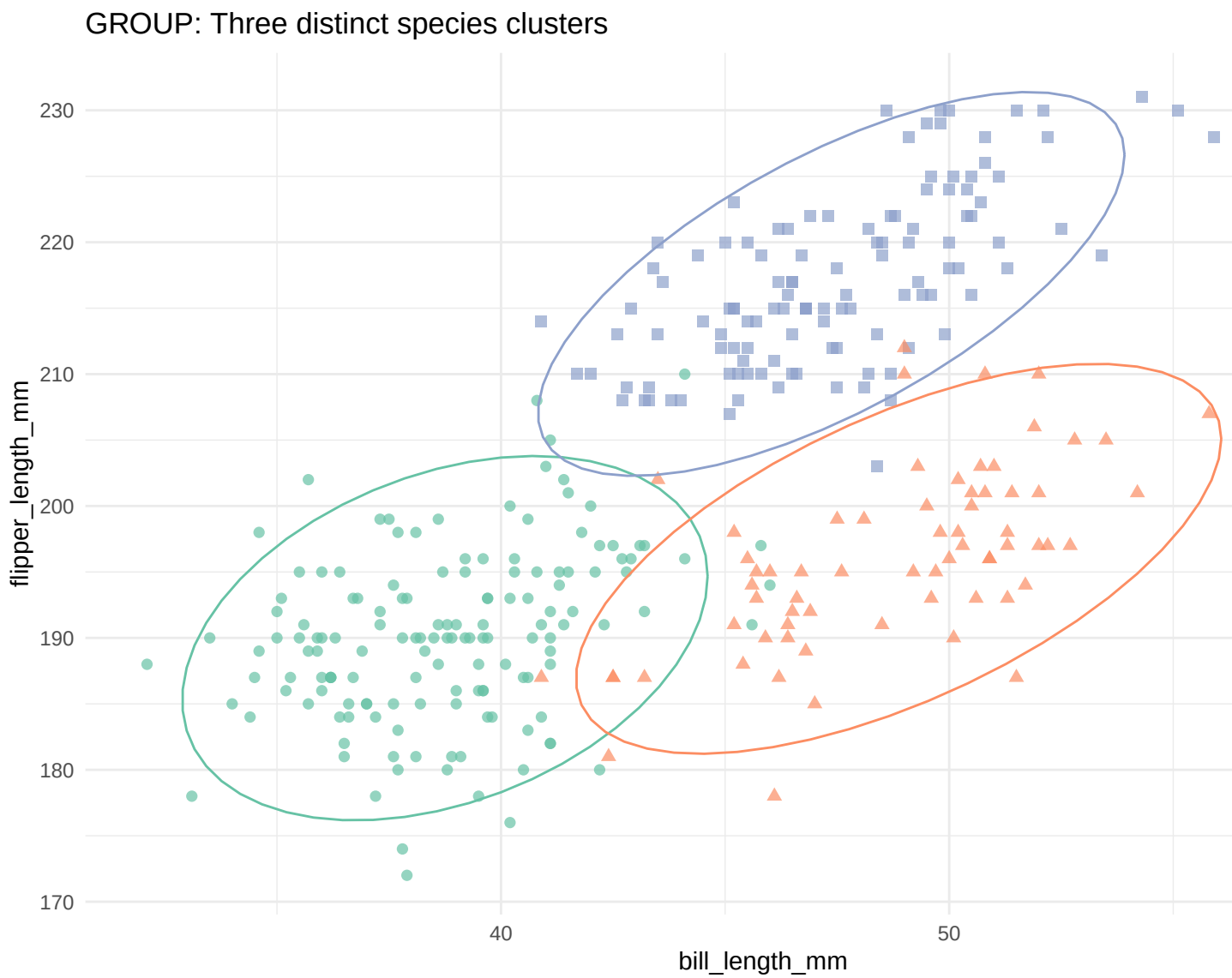


Lab 3: GEDA + Analytic Questions

```
ggplot(penguins_clean, aes(bill_length_mm, bill_depth_mm, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "CONNECT: Bill length vs depth (Simpson's paradox)") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```




```
ggplot(penguins_clean, aes(bill_length_mm, flipper_length_mm,
                           color = species, shape = species)) +
  geom_point(size = 2, alpha = 0.7) +
  stat_ellipse(level = 0.95) +
  labs(title = "GROUP: Three distinct species clusters") +
  scale_color_brewer(palette = "Set2") +
  theme_minimal()
```



Lab 4: Heat Map

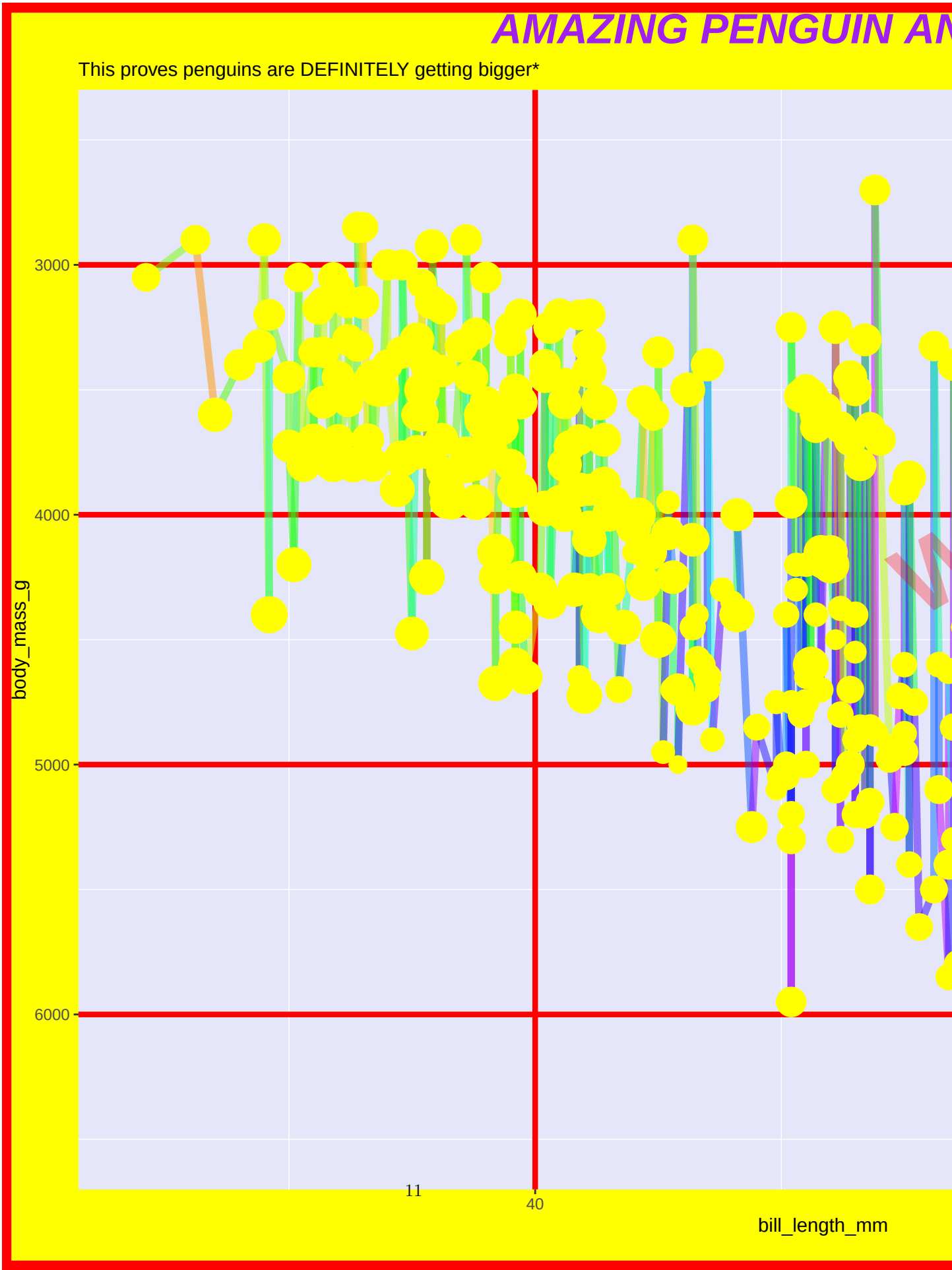
```
cor_matrix <- airquality |>
  select(Ozone, Solar.R, Wind, Temp) |>
  drop_na() |> cor()

cor_long <- cor_matrix |>
  as.data.frame() |>
  rownames_to_column("var1") |>
  pivot_longer(-var1, names_to = "var2", values_to = "correlation")

ggplot(cor_long, aes(var1, var2, fill = correlation)) +
  geom_tile(color = "white", size = 1) +
  geom_text(aes(label = round(correlation, 2)), size = 5) +
  scale_fill_gradient2(low = "steelblue", mid = "white", high = "coral",
                       midpoint = 0) +
```

```
  labs(title = "Correlation Heatmap: Air Quality") +
  theme_minimal() + theme(panel.grid = element_blank()) +
  coord_fixed()
```

Lab 5: The Ugliest Plot

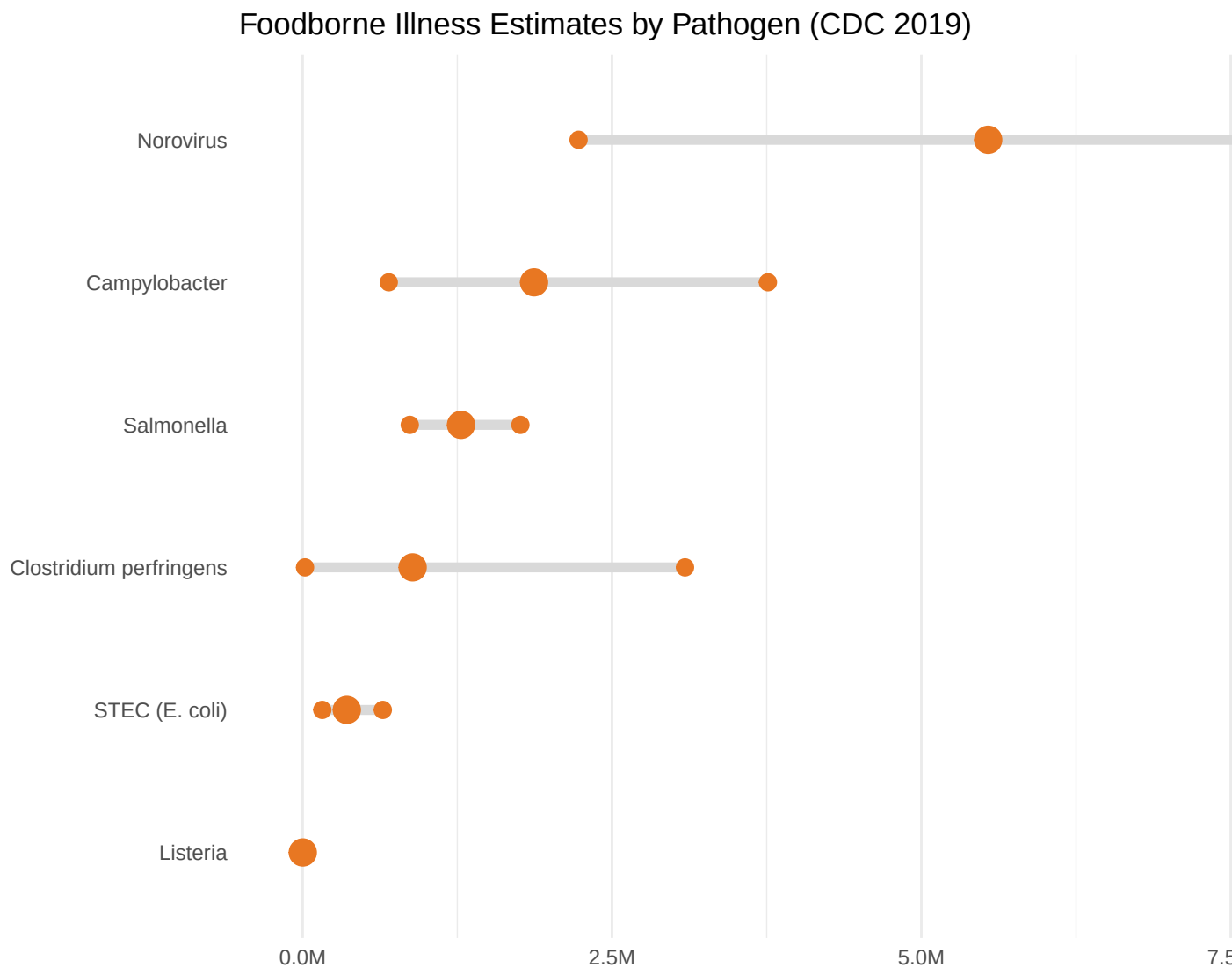
```
ggplot(penguins_clean, aes(bill_length_mm, body_mass_g)) +
  geom_line(aes(color = flipper_length_mm), size = 2, alpha = 0.5) +
  geom_point(aes(size = bill_depth_mm), color = "yellow", stroke = 3) +
  scale_color_gradientn(colors = rainbow(20)) +
  scale_y_reverse(limits = c(6500, 2500)) +
  labs(title = "AMAZING PENGUIN ANALYSIS!!!",
       subtitle = "This proves penguins are DEFINITELY getting bigger*",
       caption = "*not actually tested") +
  theme(
```

```r
  plot.background = element_rect(fill = "yellow", color = "red", size = 5),
  panel.background = element_rect(fill = "lavender"),
  panel.grid.major = element_line(color = "red", size = 1.5),
  plot.title = element_text(size = 24, color = "purple", face = "bold.italic",
                            hjust = 0.5)
) +
annotate("text", x = 50, y = 4000, label = "WOW!", size = 20,
         color = "red", alpha = 0.3, angle = 30, fontface = "bold")
```


AMAZING PENGUIN AN
This proves penguins are DEFINITELY getting bigger*
3000
4000
5000
6000
body_mass_g
40
bill_length_mm

Lab 6: Visualization Recreation

```r
pathogens <- read_csv("../data-I2Rsw.csv") |>
  rename(pathogen = Pathogen, illnesses = Illnesses,
         low = `Low Estimate`, high = `High estimate`) |>
  mutate(pathogen = fct_reorder(pathogen, illnesses))

ggplot(pathogens, aes(y = pathogen)) +
  geom_segment(aes(x = low, xend = high, yend = pathogen),
               color = "#D9D9D9", size = 2) +
  geom_point(aes(x = low), color = "#E87722", size = 3) +
  geom_point(aes(x = illnesses), color = "#E87722", size = 5) +
  geom_point(aes(x = high), color = "#E87722", size = 3) +
  scale_x_continuous(labels = scales::label_number(scale = 1e-6, suffix = "M")) +
  labs(title = "Foodborne Illness Estimates by Pathogen (CDC 2019)",
       x = "", y = "") +
  theme_minimal() + theme(panel.grid.major.y = element_blank())
```



Labs 7–9: Munzner Foundations and Color

These labs are completed as separate worksheets covering:

- **Lab 7**: Data Abstraction (What-Why-How framework)
- **Lab 8**: Marks and Channels, Expressiveness and Effectiveness
- **Lab 9**: Complexity Families, Color spaces, Color accessibility

Session Info

```r
sessionInfo()
```

```
## R version 4.1.2 (2021-11-01)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
```

```
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] lawstat_3.6          GDAdata_0.93         palmerpenguins_0.1.1
##  [4] MASS_7.3-55          patchwork_1.1.1      lubridate_1.9.5
##  [7] forcats_1.0.1        stringr_1.6.0        dplyr_1.2.0
## [10] purrr_1.2.1          readr_2.2.0          tidyr_1.3.2
## [13] tibble_3.3.1         ggplot2_3.3.6        tidyverse_2.0.0
## [16] conflicted_1.2.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1   xfun_0.56          splines_4.1.2      lattice_0.20-45
##  [5] colorspace_2.1-2   vctrs_0.7.1        generics_0.1.4     htmltools_0.5.9
##  [9] viridisLite_0.4.3  yaml_2.3.12        mgcv_1.8-39        rlang_1.1.7
## [13] pillar_1.11.1      glue_1.8.0         withr_3.0.2        bit64_4.6.0-1
## [17] RColorBrewer_1.1-3 lifecycle_1.0.5    munsell_0.5.0      gtable_0.3.6
## [21] mvtnorm_1.3-3      memoise_2.0.1      evaluate_1.0.5     labeling_0.4.3
## [25] knitr_1.51         tzdb_0.5.0         fastmap_1.2.0      parallel_4.1.2
## [29] scales_1.2.0       cachem_1.1.0       vroom_1.7.0        bit_4.6.0
## [33] farver_2.1.2       otel_0.2.0         Kendall_2.2.2      hms_1.1.4
## [37] digest_0.6.39      stringi_1.8.7      rbibutils_2.4.1    grid_4.1.2
## [41] Rdpack_2.3         cli_3.6.5          tools_4.1.2        magrittr_2.0.4
## [45] crayon_1.5.3       pkgconfig_2.0.3    Matrix_1.4-0       timechange_0.4.0
## [49] rmarkdown_2.30     R6_2.6.1           boot_1.3-28        nlme_3.1-155
## [53] compiler_4.1.2
```